

Michael Vaden

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, COLLEGE OF COMPUTING

Atlanta, GA

B.S. Computer Science

Expected: May 2026

GPA: 3.51/4.0

- **Concentrations:** Intelligence/Systems and Architecture
- **Coursework:** Machine Learning, Applied Combinatorics, Automata and Complexity, Perception and Robotics, Design and Analysis of Algorithms, Systems and Networks, Probability and Statistics, Elementary Differential Equations, Design of Operating Systems

EXPERIENCE

Delta

Atlanta, GA

Data Co-Op, Tools and Technology

September 2023 – May 2025

- Designed and deployed AWS data lake pipelines integrating S3, Athena, Glue, and Lake Formation capable of ingesting multi-TB datasets and enabling downstream analytics
- Built standardized ETL jobs improving data validation and reducing manual reconciliation by multiple hours a week
- Shipped validation service detecting duplicate invoicing and foreign exchange rate discrepancies, which detected a low 7-figure duplicate payout event and now prevents >\$5000/month routine discrepancies
- Delivered BI dashboards used by multiple diverse teams for ops decisioning

University of Georgia

Athens, GA

Software Engineering Intern

January 2022 – December 2022

- Built Django web app for research groups to manage HPC resources on the university's HPC cluster, Sapelo2
- Implemented CAS-based authentication system for secure login with university SSO
- Integrated SMTP-based notification and quota alerts improving cluster utilization
- Automated Slurm provisioning workflows reducing allocation time from a manual multi-hour job to minutes
- Coordinated with HPC admins to standardize documentation adopted by dozens of new labs and enabling more efficient onboarding and service request resolution

Georgia Tech IFC

Atlanta, GA

Web Developer

September 2024 – November 2024

- Built and deployed modern, mobile responsive website for GT IFC deployed via DigitalOcean and custom NGINX config
- Integrated admin update workflows, logging, and other server-based tools to secure the site

PROJECTS

Polygnome

Atlanta, GA

Software Developer

www.github.com/SwampPear/polygnome

- Desktop app enabling users to compose and execute structure AI tasks using composable thought chains
- Implemented with Electron serving Next.js frontend and Go Fiber for service API

Blam

Atlanta, GA

Software Developer

www.github.com/SwampPear/blam

- My experimental programming language combining features of languages I enjoy
- Authored formal EBNF grammar and built minimal LLVM-based compiler bootstrapper in C++ with clang code generation
- Roadmap includes expanded type system, native compiler, and minimal standard library

Personal Website

Atlanta, GA

Web Developer

www.github.com/SwampPear/personal-website

- Developed modern responsive site via Next.js, Tailwind, and Framer for my personal portfolio and information
- Modeled 3D objects with Blender, using Three.js for responsive animations, and implemented Perlin background via WebGL

Djinn Code Generation CLI

Atlanta, GA

Software Developer

www.github.com/SwampPear/djinn

- Rust-based CLI for automatic code generation with flexible API for natural language conversion via templating
- Implemented orchestration layer that compiles actions from natural language output

Graph of Experts Model

Atlanta, GA

Research

www.github.com/SwampPear/goe

- Ongoing research of modular mixture-of-experts architecture with expert traversal modeled as a trainable graph
- Authoring working paper formalizing model math and training objectives
- Implemented fast parallelized data download and processing scripts via thread pool executors

SKILLS

Languages: Rust, Go, C++, C, Python, JavaScript, TypeScript, SQL, Java

Libraries/Frameworks: NumPy, TensorFlow, React.js, Next.js, FastAPI, Django, LLVM, Node.js, Fiber, Selenium, OpenGL, WebGL

Technologies/Tools: Git, UI, Unix, Make/CMake, Blender, Microsoft Power Suite, Adobe Creative Suite, Tableau, Teradata

Cloud/DevOps: AWS (Bedrock, S3, Lambda, Glue, Lake Formation, Athena, Redshift), GCP (Cloud Functions, Firestore), Docker